

AMANDEEP SINGH

BACKEND DEVELOPER

Ferozpur, Punjab | +91 94648 20510 | amandeep2975@gmail.com
GitHub: <https://github.com/Aman2975>

SUMMARY

Backend Developer skilled in building scalable and secure RESTful APIs using Node.js, Express.js, Redis, and MySQL/MongoDB. Experienced in developing AI-powered applications, integrating Groq AI APIs, and building plagiarism detection systems with text analysis and intelligent explanation features. Strong understanding of authentication, database design, backend architecture, and performance optimization. Passionate about building efficient server-side applications and solving real-world problems.

SKILLS

- Languages: JavaScript (Node.js), Python
- Frameworks: Express.js, React.js
- Databases: MySQL, MongoDB, Redis
- Tools: Docker, Git, GitHub, Postman
- Technologies: REST APIs, JWT Authentication, API Security, AI API Integration,
- Concepts: Data Structures, DB Design, Error Handling, MVC Architecture, Backend Optimization, Asynchronous Processing

CERTIFICATION AND TRAINING

Full Stack Web Development Training – Department of Computer Science Punjabi University, Patiala

(MCA Industrial Training Program)

- Duration: 1 Jan 2026 – 5 May 2026
- Learned and worked on HTML, CSS, JavaScript, React.js, Node.js, Express.js, MongoDB, Python, and Redis
- Developed backend applications and AI-integrated projects during training
- Built an AI-Powered Plagiarism Detection System with Redis-based task processing and Groq AI API integration

PROJECTS

AI-Powered Plagiarism Detection System (Major Project)

- Developed a scalable plagiarism detection system using Python, Mongoose, Redis, and Node.js
- Designed and implemented a decoupled microservices architecture utilizing Node.js for high-throughput CRUD operations and FastAPI for heavy data processing
- Integrated Groq AI API for AI-generated plagiarism explanations and content analysis
- Used NLP and text-processing Python libraries for document comparison and similarity detection
- Optimized backend performance and reduced blocking operations through task queue architecture

- Designed modular and maintainable backend structure for handling large document-processing workflow

Scalable Quiz Platform (Personal Project)

- Developed RESTful APIs using Node.js and Express.js for a multi-user quiz system
- Implemented JWT-based authentication and role-based access control
- Designed and optimized MySQL database schema for better performance
- Handled errors efficiently using centralized middleware

Hugli Printing Press Website (Freelance Project)

- Developed a responsive product showcasing website for a printing press business
- Designed frontend pages for displaying products, services, and company information
- Built user-friendly UI using HTML, CSS, JavaScript, and React.js
- Focused on clean design, responsive layout, and better user experience across devices
- Worked directly on client requirements and website presentation design

EDUCATION

Master of Computer Applications (MCA)

Punjabi University – Patiala, India

Expected: May 2026

Bachelor of Computer Applications (BCA)

Punjab Technical University – Jalandhar

Completed: May 2023

- CGPA: 7.0